INCH-POUND

A-A-59372
November 19, 2003

COMMERCIAL ITEM DESCRIPTION
CALIBRATOR, OSCILLOSCOPE

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) describes an automated upgradeable oscilloscope calibration workstation with an amplitude of +/- 200 Volts DC into a 1 Meg Ω load with an accuracy of +/- 0.025% + 25 µvolts.

2. SALIENT CHARACTERISTICS. The equipment shall be capable of operation and bid sample testing, if so requested, within the accuracies, limits, and specifications herein. Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall be Class 4 in accordance with MIL-PRF-28800. This purchase description lists the minimum performance, design, functional environmental, reliability requirements and quality assurance provisions for an oscilloscope calibration system. This oscilloscope calibrator must be brand name or equal to the Fluke 9500B/1100 model oscilloscope calibrator with one 9530 150ps and 500ps pulse rise time and 3.2GHz capable active head. The AC characteristic of operation shall include leveled sine/sine to a maximum of 1.1 GHz and rise time capability to <150ps. This system shall be designed in a manner with the capability to be upgradeable to a minimum of 6.4GHz with enhanced parameters at a later date.

2.1 Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: WR-ALC/LEEC, 295 Byron Street, Robins AFB, GA 31098-1611

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2.2 Mean Time Between Failures (MTBF). The equipment Mean Time Between Failures (MTBF) shall be a minimum of 4,000 hours.

2.3 Pre-Conditioning Requirement. Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn in and thermal cycling, with thermal cycling being accomplished in the sample-testing phase.

2.4 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration and maintenance adjustments. These adjustments shall be provided by variable value components, which are adjustable, by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater, based on an operating time of 2000 hours per year.

2.5 Performance Characteristics. This instrument shall have the following salient parameters/features.

2.5.1. DC Voltage Operation.

2.5.1.1 DC Voltage Amplitude of the instrument shall have the capability of producing a DC voltage for calibration purposes of from +/1 mVDC to 200 VDC into a 1 Meg O load. Alternatively, this instrument shall have the capability of producing a DC voltage for calibration purposes of from +/1 mVDC to 5 VDC into a 50 O load.

2.5.1.2 DC Voltage Accuracy of the instrument shall be an accuracy of +/- 0.025% + 25 µvolts.

2.5.1.3 DC Voltage Ranging of the instrument shall have a DC voltage range of 1, 2, 5 or 1, 2, 2.5, 4, 5 or a continuous integers rating.

2.5.1.4 DC Voltage Deviation shall be +/- 11.2%.

2.6 Squarewave Output Parameters

2.6.1 Squarewave Output Amplitude. This instrument shall have an amplitude range of 40 µvolts to 200 Volts p-p into a 1 Meg O load or 40 µvolts to 5 Volts p-p into a 50 O load. The squarewave's polarity may be positive, negative or symmetrical about ground. The ranging shall be the same as paragraph 2.5.1.3 but for a squarewave output, with a deviation also equal to that stated in paragraph 2.5.1.4.

2.6.2 Squarewave Output Rise/Fall Time. This instrument shall have a squarewave rise/fall time below 100 volts or below of < 150 nanoseconds, or have a squarewave rise/fall time at >= 100 volts of < 200 nanoseconds.

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2.6.2.1 Squarewave Output Aberrations. This instrument shall have a squarewave aberration <2% peak for first 500ns; <0.1% after 500ns; and <0.01% after 100us.".

2.6.2.2 Squarewave Output Frequency. This instrument shall have a squarewave output frequency range of 10 Hz to 100 kHz at an accuracy of +/- 0.25 ppm with ranging the same as that stated in paragraph 2.5.1.3.

2.7 Low-Edge Pulse Parameters.

2.7.1 Low-Edge Pulse Amplitude shall be an amplitude range of 5 millivolts to 3 Volts p-p into a 50 O load with an accuracy of + 50 picoseconds to – 150 picoseconds. The ranging shall be the same as that stated in paragraph 2.5.1.3.

2.7.2 Low-Edge Pulse Rise/Fall Time of the instrument shall have a rise/fall time on the low-edge pulse of 150 picoseconds return to ground.

2.7.3 Low-Edge Pulse Mark/Space Ratio shall have a low-edge pulse mark-to-space ratio of 1:9.

2.7.4 Low-Edge Pulse Aberration into a VSWR of 1.2 : 1 This instrument shall have a pulse aberration at this stated VSWR of < +/- 2% peak in 8 GHz and < +/- 1.5% peak in 3 GHz for the first 10 nanoseconds.

2.7.5 Low-Edge Pulse Frequency This instrument shall have a frequency range of on this type of signal 10 Hz to 2 MHz with an accuracy of +/- 0.25 ppm with ranging that shall be the same as that stated in paragraph 2.5.1.3.

2.8 High-Edge Pulse Parameters.

2.8.1 High-Edge Pulse Amplitude shall be an amplitude range of 1 millivolts to 200 Volts p-p into a 1 MO load or 1 millivolts to 5 Volts p-p into a 50 O load with an accuracy of +/- 3%. The ranging shall be the same as that stated in paragraph 2.5.1.3.

2.8.2 High-Edge Pulse Rise/Fall Time This instrument shall have a rise/fall time on the high-edge pulse of an amplitude of < 100 volts at < 150 nanoseconds or > or = 100 volts at < 200 nanoseconds return to ground.

2.8.3 High-Edge Pulse Mark/Space Ratio This instrument shall have a high-edge pulse mark-to-space ratio of 1:1.

2.8.4 High-Edge Pulse Aberration shall be a pulse aberration of < +/- 2% peak for the first 500 nanoseconds.

2.8.5 High-Edge Pulse Frequency shall be a frequency range of on this type of signal 10 Hz to 100 kHz with an accuracy of +/- 0.25 ppm with ranging that shall be the same as that stated in paragraph 2.5.1.3.

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2.9 Timing Markers. The following parameters apply to timing markers.

2.9.1 Timing Markers Styles. There shall be the choice of three types, as follows: 1) Square/Sinewave 2) Pulse or 3) A Narrow Triangle.

2.9.2 Timing Markers Square/Sinewave. The period on a squarewave marker shall be from 10 nanoseconds to 55 seconds. The period on a sine wave marker shall be from 450 picoseconds to 9.009 nanoseconds.

2.9.3 Timing Markers Pulsed. The period on a pulsed timing marker shall be from 1.0 microseconds to 55 seconds, with a rise time and fall time of typically 1 nanosecond.

2.9.4 Timing Markers Narrow Triangle. The period on a narrow triangle timing marker shall be from 1.0 microseconds to 55 seconds, with a rise time and fall time of typically 2.5% of the period.

2.9.5 Timing Markers Ranging shall be 1, 2, 5 or 1, 2.5, 4, 5 or continuous for a period of all of the wave shapes listed in paragraph 2.9.1. of this CID.

2.9.6 Timing Markers Timing Jitter shall be less than 10 picoseconds pk-pk with time window less than 100 nanoseconds, less than 100 picoseconds pk-pk with time window less than 100 microseconds and less than 1000 picoseconds pk-pk with time window less than 10 milliseconds.

2.9.7 Timing Markers Deviation shall be +/- 45% for the period.

2.9.8 Timing Markers Amplitude shall be 100 mV to 1 Volt peak-to-peak.

2.9.9 Timing Markers Sub-Division shall be such that every tenth marker can be set to higher amplitude for period +/- 1 µseconds for all waveforms.

2.9.10 Timing Markers Accuracy Nominally, this parameter shall be +/- 0.25 ppm.

2.10 Leveled Sine Function Parameters.

2.10.1 Leveled Sine Function Frequency Range shall be 0.1 Hz to 1.1 GHz.

2.10.2 Leveled Sine Function Amplitude Peak-to-Peak into a 50 O Load shall be the following: From 0.1 Hz to 550 MHz shall be 5 mV to 5 Volts. From 550 MHz to 1.1 GHz shall be 5 mV to 3 Volts.

2.10.3 Leveled Sine Function Accuracy shall be +/- 1.5% at a single reference rate of 50 kHz to 10 MHz.

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2.10.4 Leveled Sine Function Flatness Reference Frequency Into VSWR of 1.6:1 (1.2:1) shall be 0.1 Hz to 300 MHz +/- 2%. From 300 MHz to 550 MHz this shall be 3% (+/- 2.5%), from 550 MHz to 1 GHz this shall be 4% (+/- 3.5%).

2.10.5 Leveled Sine Function Harmonic Purity shall be less than -35 dBc on the second harmonic and less than -40 dBc at the third harmonic in 12 GHz.

2.10.6 Leveled Sine Function Non- and Sub- Harmonic Purity shall be less than -40 dBc.

2.11 Dual Sine Function Parameters.

2.11.1 Dual Sine Function Frequency Range shall be 0.1 Hz to 1.1 GHz Note: Not relevant for single head configurations.

2.11.2 Capacitance Measurement Versus Accuracy shall be as follows: From 1 pF to 35 pF this shall be +/- 2% +/- 0.25 pF and from 35 pF to 95 pF shall be +/- 3% +/- 0.25 pF.

2.12 Pulse Width Function Parameters (These shall not be available with use of an active head)

2.12.1 Pulse Width shall be 1 nanosecond to 100 nanoseconds.

2.12.2 Pulse Width Accuracy shall be less than 5% +/- 200 picoseconds.

2.12.3 Pulse Width Adjustment Resolution shall have the following criteria: less than 50 picoseconds from 1 nanoseconds to 4 nanoseconds; less than 250 picoseconds from 4 nanoseconds to 20 nanoseconds; and less than 1 nanoseconds from 20 nanoseconds to 200 nanoseconds.

2.12.4 Pulse Width Rise and Fall Times shall be less than 450 picoseconds.

2.12.5 Pulse Width Aberrations shall be less than +/- 5% peak typical.

2.12.6 Pulse Width Stability shall be less than 10 picoseconds peak-to-peak over 10 minutes per degree Celsius.

2.12.7 Pulse Width Jitter (Trigger) This shall be less than 5 picoseconds peak-to-peak.

2.12.8 Pulse Width Frequency This shall be 1 kHz to 1 MHz.

2.12.9 Pulse Width Amplitude shall be 1 Volt peak-to-peak into 50 O.

2.13 Other Output Options– Parameters.

2.13.1 Magnitude of DC Current shall be +/- 100 µA to +/- 100 mA.

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2.13.2 Accuracy of DC Current shall be (0.25% + 5 µA).

2.13.3 Duty Cycle Symmetry of Current Squarewaves shall be 50% centered about ground.

2.13.4 Rise Time and Aberrations of Current Squarewaves shall be less than 150 nanoseconds and less than 2% peak.

2.14 Composite Video Parameters.

2.14.1 Composite Video Amplitude shall be1.0 volt peak-to-peak.

2.14.2 Composite Video Pattern (Full Raster) shall be white, gray or black.

2.14.3 Composite Video Sync Polarity shall be positive or negative.

2.14.4 Composite Video Standards shall be 625-lines 50 Hz or 525-lines 60 Hz.

2.14.5 Composite Video Trigger Output shall be composite sync or odd field starting.

2.15 Signal Routing shall be through an SMA connector mounted on the rear panel as a passive input and also a switched 50 O path to any active head that may be attached.

2.16 Maximum Input Signal shall be up to +/- 40 volts peak-to-peak, or +/- 400 mA peak-to-peak.

2.17 Insertion Loss into 50 O shall be less than 2.5 dB up to 100 MHz, less than 4 dB up to 500 MHz, and less than 6.0 dB up to 1.0 GHz.

2.18 Reference Frequency Ranges shall be as follows: 1) 1 MHz to 20 MHz in 1 MHz steps at the BNC input and 2) 1 MHz or 10 MHz at the BNC output.

2.19 Lock Range of Reference Frequency shall be +/- 50 ppm.

2.20 Environmental Associated Parameters.

2.20.1 Operating Temperature shall be 5 °C to 40 °C.

2.20.2 Non-Operating Temperature shall be 0 °C to 50 °C.

2.20.3 Operating Humidity Range shall be at less than 90% RH - 5 °C to 30 °C and non-condensing less than 75% RH 30 °C to 40 °C.

2.20.4 Non-Operating Humidity Range shall be at less than 95% RH - 0 °C to 50 °C.

2.20.5 AC Line Voltage shall be 95 volts to 132 volts rms or 209 volts to 264 volts rms.

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2.20.6 AC Line Frequency shall be 48 Hz to 63 Hz.

2.20.7 Power consumption shall be equal to or less than 400 Volt-Amperes.

2.20.8 Warm up time shall be 20 minutes.

2.21 Dimensions and Weight. Size shall be no greater than the H x W x D of 5.5 inches, 17 inches and 17.5 inches, respectively. Total weight shall be no greater than 30 pounds.

2.22 Operator, maintenance, and illustrated parts manuals shall be supplied at no additional cost. Format and quantity shall be as specified in the contract or order.

2.23 Warranty period shall be one year.

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR). These are consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials.

4. PRODUCT CONFORMANCE PROVISIONS

4.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance before first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of ASTM-SI10, and all other requirements of this CID are met.

4.3 Exceeded Metric Products. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order. The Equipment shall have an identification plate permanently affixed in a readable place, which includes the information contained in MIL-PRF-28800, Section 3.

6. NOTES.

6.1 Sources of documents.

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6.1.1 Government Documents. Military Specifications, Standards, and Handbooks referenced herein may be obtained from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

6.1.2 Code of Federal Regulations (CRF). Copies of the CFR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

6.1.3 ASTM standards. Copies of ASTM standards can be obtained from ASTM, 100 Bar Harbor Drive, West Conshohocken, PA 19428-2959. Phone 1-610-832-9500. Fax 1-610-832-9555.

MILITARY INTEREST

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